Ans. (d) : Taken time by 'w' for complete the work

$$\Rightarrow \frac{30}{25}\times 100$$

$$\Rightarrow 120 \text{ days}$$

taken time by 'x' for complete the work

$$= 10\times\frac{4}{1} = 40 \text{ days}$$

taken time by 'y' for complete the work

$$= \frac{40}{40}\times 100 = 100 \text{ days}$$

taken time by 'z' for complete the work

$$= 13\times\frac{3}{1}$$

$$= 39 \text{ days}$$

$\because$ Efficiency $\propto \frac{1}{\text{time}}$

Hence, Z will complete the work first.

77. The average monthly income of a person in a certain family of 5 is Rs. 10,000 . What will be the average monthly income of a person in the same family if the income of one person increased by Rs. 1,20,000 per year ?

(a) Rs 12,000
(b) Rs 16,000
(c) Rs 20,000
(d) Rs 34,000

Ans. (a) :

$\because$ Average monthly income of family of 5 members = 10,000 ₹

$\therefore$ Total monthly income = 10,000 × 5
= 50,000 ₹

$\because$ The income of one person increased by 120,000 per year.

$\therefore$ Monthly increased income of that person

$$= \frac{1,20,000}{12} = ₹\ 10,000$$

Total income per month = 50,000 + 10,000
= 60,000

$\therefore$ Required average $= \frac{60,000}{5}$
= 12,000

OR

$$\text{Required Average} = 10,000 + \left(\frac{1,20000}{12}\right)\times\frac{1}{5}$$

$$= 10,000 + (10,000)\ \frac{1}{5}$$

$$= 10,000 + 2000$$

$$= 12000$$

78. In a race, a competitor has to collect 6 apples which are kept in a straight line on a track and a bucket is placed at the beginning of he track which is a starting point. The condition is that the competitor can pick only one apple at a time, run back with it and drop it in the bucket. If he has to drop all the apples in the bucket, how much total distance he has to run if the bucket is 5 meters from the first apple and all other apples are placed 3 meters apart?

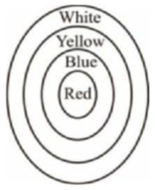
(a) 40 m (b) 50 m
(c) 75 m (d) 150 m

Ans. (d) : The distance covered by competitor
= 2(5+8+11+14+17+20) = 2 × 75 = 150 m.

79. A round archery target of diameter 1 m is marked with four scoring regions from the centre outwards as red, blue, yellow and white. The radius of the red band is 0.2m. The width of all the remaining bands is equal. If archers throw arrows towards the target, what is the probability, that the arrows fall in the red region of the archery target ?

(a) 0.40 (b) 0.20
(c) 0.16 (d) 0.04

Ans. (c) :



Diameter of archery target = 1m
Radius of red Band = .2m
Area of circle = πR^2

$$= \pi.(.2)^2 = 0.04\pi = \frac{22}{7}\times 0.04 = \frac{88}{700}\ m^2$$

$$\text{Total area} = \frac{22}{7}\times.5\times.5 \text{ or } \frac{22}{7}\times\frac{5}{10}\times\frac{5}{10}$$

$$= \frac{22}{7\times 4} = \frac{11}{14}\ m^2$$

$$\text{Required probability} = \frac{\text{Area of red}}{\text{Total Area}} = \frac{\frac{88}{700}}{\frac{11}{14}}$$

$$= \frac{88}{700}\times\frac{14}{11} = 0.16$$

80. A person allows 10% discount for cash payment from the marked price of a toy and still he makes a 10% gain. What is the cost price of the toy which is marked Rs.770 ?

(a) Rs. 610 (b) Rs. 620
(c) Rs. 630 (d) Rs. 640

Ans. (c) : Marked Price of Toy = Rs 770
Discount % = 10%
Formula :
Marked Price:cost price=100+Profits%:100 – Discount
Marked Price : Cost Price = 110 : 90 = 11:9
Marked Price =770
11x = 770
x = 7
Cost Price of Toy = 9x = 9×70 = 630

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2017

CSAT : PAPER-II

(Chapterwise Analysis with Explanation)

Time : 2 hours **Maximum Number : 200**

COMPREHENSION

Directions for the following 8 (eight) items :
Read the Following eight passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

What climate change will undeniably do is cause of amplify events that hasten the reduction of resources. Competition over these diminising resources would ensue in the form of political or even violent conflict. Resource based conflicts have rarely been overt and are thus difficult to isolate. Instead they take on veneers that appear more politically palatable. Conflicts over resources like water are often cloaked in the guise of identity or ideology.

1. What does the above passage imply?

(a) Resource-based conflicts are always politically motivated.
(b) There are no political solutions to resolve environmental and resource based conflicts.
(c) Environmental issues contrilaite to resource stresses and politic ril conflict.
(d) Political conflict based on identity or ideology cannot be resolved.

Ans. (c) : Indiscriminate industrialization and urbanization have led to excessive exploitation of natural resources. The problem of climate change has increased this complexity and has created a competition for supremacy over natural resources. The political leadership continues to generate political struggle for the hegemony of natural resources by promoting these issues.

Passage-2

The man who is perpetually hesitating which of the two things he will do first, will do neither. The man who resolves, but suffers his resolution to be changed by the first counter Suggestion of a friend—who fluctuates from opinion to opinion and veers from plan to plan-can never accomplish anything. He will at best be stationary and probably retrograde in all. It is only the man who first consults wisely, then resolves firmly and then executes his purpose with inflexible perseverance, undismayed by those petty difficulties which daunt a weaker spirit—that can advance to eminence in any line.

2. The keynote that seems to be emerging from the passage is that

(a) we should first consult wisely and then resolve firmly
(b) we should reject suggestions of friends and remain unchanged
(c) we should always remain broad-minded
(d) we should be resolute and achievement-oriented

Ans. (a) : The exact keynote that emerges from the passage, we can see that the author is talking about to consult wisely and to resolve firmly with perseverance in the absence of flexibility.

Passage-3

During the summer in the Arctic Ocean, sea ice ha, been melting earlier and faster, and the winter freeze has been coming later. In the last three decades, the extent of summer ice has declined by about 30 per cent. The lengthening period of summer melt threatens to undermine the whole Arctic food web, atop which stand polar bears.

3. Which among the following is the most crucial message conveyed by the above passage?

(a) Climate change has caused Arctic summer to be short but temperature to be high.
(b) Polar bears can be shifted to South Pole to ensure their survival.
(c) Without the presence of polar bears, the food chains in Arctic region will disappear.
(d) Climate change poses a threat to the survival of polar bears.

Ans. (d) : The problem of global warming has intensified climate change. This has put a threat to biodiversity, Due to global warming, the glaciers of the arctic are melting, due to which there has been a serious threat to the flora and fauna of the Arctic. Polar bear have been affected the most by the crisis. Polar bears are listed as endangered in the IUCN Red Data Book.

Passage-4

Why do people prefer open defecation and not want toilets or, if they have them, only use them sometimes? Recent research has shown two critical elements: ideas of purity and pollutions, and not wanting pits or septic tanks to fill they have to be emptied. These are the issue that nobody wants to talk about, but if we want to eradicate the practice of open defection, they have to be confronted and dealt properly.

4. Which among the following is the most crucial message conveyed by the above passage?

(a) The ideas of purity and pollutions are so deep-rooted that they cannot be removed from the minds of the people.
(b) People have to perceive toilet use and pit-emptying as clean and not polluting.